

Name _____

Section _____

MTH:2310 DISCRETE MATH
THE LAST QUIZ: QUIZ 10
DUE: WEDNESDAY DEC. 10

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)

Your ability to express your solutions is just as important as mathematical accuracy.

You will be penalized up to 4 points for notation and disorganized or hard to read solutions.

Problem 1. (12 points) Relations and Their Properties

- (a) (8 points) Let $R = \{(c, b), (a, c), (c, c), (a, b), (b, c)\}$ be a relation on the set $A = \{a, b, c\}$. Determine whether R is reflexive and/or symmetric and/or antisymmetric and/or transitive. Justify your answer.

Not Reflexive $(a, a) \notin R$

Not Symmetric $(a, c) \in R \wedge (c, a) \notin R$

Not transitive $(b, c) \wedge (c, b) \rightarrow (b, b)$ but $(b, b) \notin R$

Not Antisymmetric $(c, b) \wedge (b, c)$

- (b) (4 points) Define a relation R' on the same set A using roster notation such that R' has at least 2 elements and is reflexive and transitive, but not symmetric.

$$R' = \{ (a, a), (b, b), (c, c), (a, b) \}$$

trivially transitive, reflexive, not symmetric

Problem 2. (12 points) Equivalence Relations

Let R be the relation on the set of all bit strings defined so that for any bit strings x and y , xRy if and only if x and y contain the same number of 1's.

(a) (8 points) Show that R is an equivalence relation.

Equality is a symmetric, transitive, and reflexive operation.

For R , we are evaluating the equality

of the number of 1's contained

as this relation is, in essence, an equality operation; and we are evaluating on the entire set of bit strings, which in this case can be represented by $\mathbb{Z}$.

(b) (4 points) Describe in words the equivalence class $[001100]_R$, then list 5 of its members.

$[001100]_R$ represents any bit string with exactly two 1's.

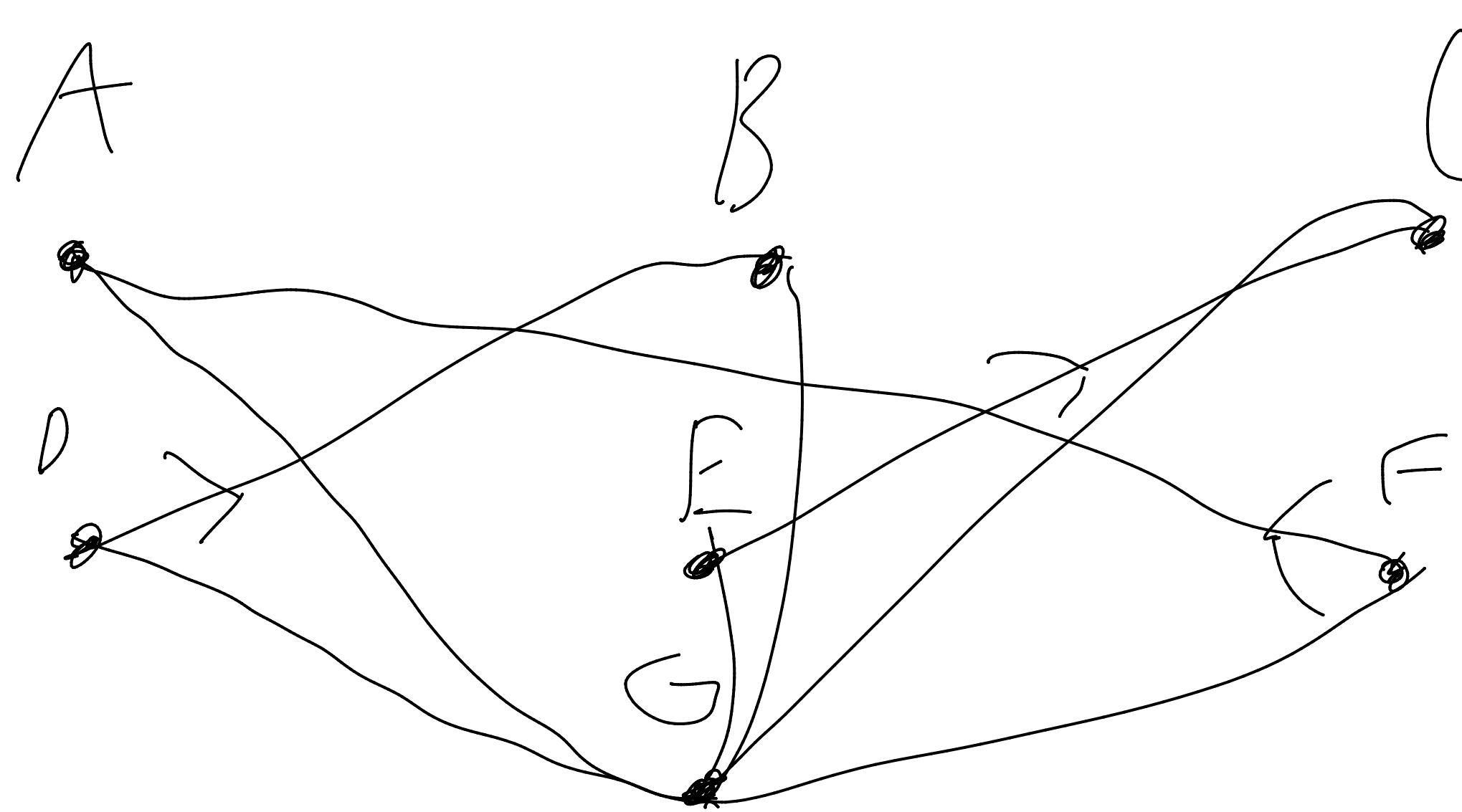
$$[001100]_R = \{11, 110, 1100, 11000, 110000\}$$

Problem 3. (20 points) *Graphs and Graph Terminology*

Consider the following sets.

$$\begin{array}{llll}
 A = \{x, y, z\} & C = \{a, b, c\} & E = \{a, b\} & G = \{z, 3, a\} \\
 B = \{1, 2, 3\} & D = \{1, 3\} & F = \{y, z\} &
 \end{array}$$

- (a) (8 points) Draw a graph where the nodes labeled as the sets A, B, C, D, E, F , or G . Connect nodes with an edge if the corresponding sets have a non-empty intersection. Include a direction to all edges where the initial node is a proper subset of the terminal node.



- (b) (3 points) What kind of graph is this (*simple, multi, pseudo, directed, mixed, undirected*)?

this is a simple mixed graph

Name _____

Section _____

(c) (4 points) Find the degree of each node of the underlying undirected graph.

$$\deg(A) = 2$$

$$\deg(E) = 2$$

$$\deg(B) = 2$$

$$\deg(F) = 2$$

$$\deg(C) = 2$$

$$\deg(G) = 6$$

$$\deg(D) = 2$$

(d) (2 points) Find all elements of $N(E)$.

$$N(E) = \{C, G\}$$

(e) (3 points) Find all elements of $N(\{A, E, G\})$.

$$N(\{A, E, G\}) = \{A, B, C, D, E, F, G\}$$

